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Experimental and kinetic modeling study of 2-butanol pyrolysis and combustion

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ABSTRACT

2-Butanol (sC_4H_9OH) pyrolysis has been studied in a flow reactor with the synchrotron vacuum ultraviolet photoionization mass spectrometry combined with the molecular-beam sampling technique. The pyrolysis species were identified and their mole fractions were determined. Four pressures of 5, 30, 150 and 760 Torr were selected to study the pressure dependence of 2-butanol pyrolysis chemistry. The temperature- and pressure-dependent rate constants of unimolecular reactions of 2-butanol were calculated with the RRKM/Master Equation method. With the help of theoretical calculations, a detailed kinetic model consisting of 160 species and 1038 reactions was developed to simulate the 2-butanol pyrolysis. It is concluded that the mole fractions of pyrolysis species are very sensitive to the 2-butanol unimolecular reaction rates. To enhance the accuracy, the model is further validated by the species profiles in shock tube pyrolysis, a rich laminar premixed flame, oxidation data from jet-stirred reactor, ignition delay times, and laminar flame speed. Good agreements between the predicted and measured results were obtained.

1. Introduction

Bio-butanol is considered to be a potential fuel due to the feasibility of producing butanol from biomass feedstock. Furthermore, bio-butanol is environmental-friendly because its net CO_2 emission associated with combustion is comparatively lower than that from fossil fuels. 2-Butanol (sC_4H_9OH), which is one of the four isomers of butanol, can be produced from glucose through the bacterial fermentation and chemical conversion with high conversion efficiency [1]. Christensen et al. studied the chemical and physical properties of several renewable oxygenates blending in gasoline [2]. The vapor pressure, vapor lock protection, distillation, density, octane rating, viscosity and the other properties of 2-butanol were measured. 2-Butanol is suggested to be one of the potential fuels for its favorable properties [2].

Compared with the extensive studies on *n*-butanol combustion [3–5], only a few studies have been performed on the combustion chemistry of 2-butanol. Gu et al. [6] and Veloo and Egolfopoulos [7] have measured the laminar burning velocities of butanol isomer–air mixtures. Moss et al. [8], Stranic et al. [9] and Yasunaga et al. [10] have studied the ignition delay times of 2-butanol in shock tube over wide ranges of temperature, pressure, and equivalence ratio. Togbé et al. [11] investigated the 2-butanol oxidation in a jet-stirred reactor over the temperature range of 770–1250 K at 10 atm and with equivalence ratios of 0.5–4.0. Concentration profiles of oxidation species were measured by gas chromatography and infrared spectrometry. Yang et al. [12] identified the intermediate species in rich premixed flames of four butanol isomers with the synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Oβwald et al. [13] investigated the chemical structures of rich premixed flames of four butanol isomers (ϕ = 1.7) at 30 Torr with SVUV-PIMS and the electron-impact ionization mass spectrometry (EIMS). More recently, Rosado-Reyes and Tsang [14] measured the unimolecular reaction rate constants of 2-butanol using the single pulse shock tube technique. Stranic et al. [15] measured the species time history profiles during 2-butanol pyrolysis at 1.3–1.9 atm with temperature range of 1270–1640 K behind shock waves. Base on the experimental results stated above, several kinetic models have been proposed recently [8,16,17].

In this work, the pyrolysis of 2-butanol in a flow reactor at pressures of 5, 30, 150, and 760 Torr were studied to understand the high temperature chemistry of 2-butanol. A kinetic model consisting of 160 species and 1038 reactions was developed to simulate the 2-butanol pyrolysis at various pressures. To enhance the

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accuracy of the model, the rate constants of dominant unimolecular decomposition channels of 2-butanol and β-scission of 2-butanol radicals (C_4H_8OH) were calculated with the RRKM/Master Equation method. To develop a 2-butanol mechanism over a wide range of temperature, pressure, and equivalence ratio, the species profiles in shock tube pyrolysis [15], a rich laminar premixed flame [13], the oxidation data in JSR [11], ignition delay times [9,10], and laminar flame speeds [7] were also used to validate the present model. The simulated results of the present model are also compared to the previous models [8,16,17].

2. Experimental method

The experiments were carried out at the National Synchrotron Radiation Laboratory in Hefei, China. Detailed descriptions of the two beamlines and pyrolysis apparatus used in this work have been published elsewhere [3,18–20]. The schematic pyrolysis apparatus is shown in Fig. S1a in the Supplementary materials. The apparatus consists of three chambers: a pyrolysis chamber, a differentially pumped chamber, and a photoionization chamber with a home-made reflectron time-of-flight mass spectrometer (RTOF-MS). In the pyrolysis chamber, there is a 6.8-mm-ID (inner diameter) alumina flow tube with 150 mm heated by a furnace. The mixture of Ar (970 standard cubic centimeter per minute (SCCM)) and 2-butanol (30 SCCM in gas phase) was fed into the flow tube. After that, the pyrolysis species were sampled at 10 mm downstream the outlet by a quartz cone-like nozzle with a 40° cone angle. Several nozzles with different orifices (70–500 μm) were used for experiments under different pressures. The sampled pyrolysis species formed a molecular beam in the differentially pumped chamber, and passed into the photoionization chamber through a nickel skimmer. The molecular beam was crossed by the tunable VUV light in the photoionization chamber and the ionized species were detected by RTOF-MS.

The temperature distributions along the centerline of the flow tube were measured by a thermocouple before and after the pyrolysis experiment. Detailed description of the temperature measurement method has been provided in previous studies [3,18]. The measured temperature profiles are illustrated in Fig. S1b in the Supplementary materials. Each profile is named by its maximum temperature (T_{max}) which is used as experimental temperature. The uncertainty of T_{max} is estimated to be within ±30 K.

The pressure at the outlet of the flow tube (P_{outlet}) can be changed from 2 to 760 Torr to study the pyrolysis under various pressures. In this work, four pressures of 5, 30, 150, 760 Torr were chosen. The Reynolds number (R_e) of the flow was calculated to be less than 2000 over temperature range from T_{max} = 600 to 1900 K and P_{outlet} = 2 to 760 Torr. Thus it is a laminar flow under the experimental conditions. The pressure distributions along the flow tube are evaluated by using (E1).

$$v\mathrm{d}v + \mathrm{d}p/\rho + \mathrm{d}F = 0 \tag{E1}$$

where v, p, ρ, and F are velocity, pressure, density and frictional resistance, respectively. The detailed calculations can be found in previous work [3,18]. When the pressure is below 30 Torr, the pressure profiles should be used in the simulation. The pressure profiles at 5 Torr with various temperatures are provided in the Supplementary materials. For the pressures above 30 Torr, the difference between inlet and outlet pressures is less than 10%, so the pressure can be treated as constant.Two experimental modes were used in this work including the measurement of photoionization efficiency (PIE) spectra to identify the pyrolysis products and the temperature scan in order to plot the mole fraction profiles of the pyrolysis species versus T_{max}. In this work, the photon energies of 16.64, 11.70, 11.00, 10.50, 10.00, and 9.50 eV were selected to obtain the near-threshold photoionization mass spectra of pyrolysis species. The method for evaluating mole fractions has been reported in detail elsewhere [21]. The photoionization cross sections (PICSs) of 2-butanol are from our previous study [22], which are available in our online database [23] along with the PICSs of other pyrolysis products. The experimental uncertainties of mole fractions are evaluated to be ±25% for products with known PICSs and a factor of 2 for those with estimated PICSs. In the present work, only the PICS of $C_2H_5COCH_3$ is unknown, which is estimated from that of CH_3COCH_3. The experimental uncertainties of C, H, and O balances are all within ±15% compared with inlet fluxes of C, H, and O elements (see Fig. S2 in the Supplementary materials).

3. Theoretical calculations and kinetic model

3.1. Theoretical methods

3.1.1. Ab initio calculations

The B3LYP/6-311++G(d,p) method was used to optimize the structures of all the stationary points involved in the reaction pathways of 2-butanol. Further frequency calculations at the same level were performed to get the zero-point energy (ZPE) and to identify each point to be the local minima or transition state (TS). The single point energies were corrected with the QCISD(T) method combined with two large basis sets: the correlation-consistent, polarized-valence, triple-ζ (cc-pVTZ), and quadruple-ζ (cc-pVQZ) basis sets of Dunning [24,25]. Then the QCISD(T) energies were extrapolated to the complete basis set (CBS) limit via the following expression [25],

$$\begin{aligned} E[\mathrm{QCISD}(T)/\infty] \approx\ & E[\mathrm{QCISD(T)/cc\text{-}pVQZ}] \\ & + E[\mathrm{QCISD(T)/cc\text{-}pVQZ}] \\ & - E[\mathrm{QCISD(T)/cc\text{-}pVTZ}] \times 0.6938 \end{aligned} \tag{E2}$$

For the direct C—C bond fission reactions without apparent barriers, the minimum energy curves were constructed using the MP2-CAS(2,2)/6-311G(d,p)//CASSCF(2,2)/6-311G(d,p) method, scanning along the dissociation bond length with the 0.2 Å intervals. The active space was chosen as the bonding and anti-bonding orbitals for the breaking C—C bonds. Morse potential $V(R) = D_e\{1 - \exp[-\beta(R - R_e)]\}^2$ was used to fit the potential curve along the individual reaction coordinate, where D_e is the electronic binding energy with the ZPE correction, and R_e is the equilibrium bond length. Various electronic structure methods other than QCISD(T) including G3B3, CCSD and CBS-APNO have been employed to examine the accuracy of the computed energies, which are listed in Table 1. All of ab initio calculations were performed with the Gaussian 09 program [26].

3.1.2. Rate constant calculations

The temperature- and pressure-dependent rate constants for the major decomposition channels were calculated by the RRKM/Master Equation method using the Variflex code [27]. Partition functions are evaluated at the E/J (energy E and total angular momentum J) resolved calculation. The step-size is chosen to be 100 cm^{-1} for E and 10 for J, respectively, which is consistent with our previous studies [3,28,29]. One-dimensional (1D) master equa-

Table 1
Bond dissociation energies (BDE_{298}, kcal/mol) of 2-butanol at various levels.

C—C bonds	QCISD(T)	G3B3	CBS-APNO	CBS-QB3[42]
$CH_3CH_2CHOH—CH_3$	87.40	87.06	87.83	88.4
$CH_3CH_2—CHOHCH_3$	85.86	86.07	86.52	86.9
$CH_3—CH_2CHOHCH_3$	90.75	90.16	91.62	91.0

tion was used to simulate the pressure-dependent rate constants. In this work, inert gas Ar is used as the bath gas. The interaction between the reactant and the bath gas was approximated by the Lennard-Jones (L-J) potential. The L-J pairwise parameters of Ar are chosen as σ = 3.465 Å, ε/k = 113.5 K [30]. For sC_4H_9OH, the L-J parameters are estimated by the following equations [31]:

$$\sigma = 2.44(T_c/P_c)^{1/3} \tag{E3}$$

$$\varepsilon_A/k_b = 0.77T_c \tag{E4}$$

where k_b is the Boltzmann constant, T_c is the critical temperature and P_c is the critical pressure. The values of T_c = 536.05 K and P_c = 41.79 bar were taken from the textbook [32]. Thereby the L-J parameters for sC_4H_9OH were computed to be σ = 5.712 Å, ε/k = 412.76 K according to (E3) and (E4). The collision energy transfer was treated using an exponential-down model with $\langle \Delta E_{down} \rangle = 150 \times (T/300)^{0.85}$ cm^{-1} (temperature unit: K), referring to previous studies [33,34]. In the rate constant calculations, low-frequency vibrational modes corresponding to internal rotations were treated as hindered rotors with computing the hindrance potentials at the B3LYP/6-31G(d) level of theory [35]. Reactions with tight transition states were treated with the conventional transition state theory [36], while the number of states was evaluated by the rigid-rotor harmonic-oscillator (RRHO) assumption. Furthermore, the rate constants were corrected by the quantum tunneling effect with the Eckart model [37]. For the barrierless reactions, the rate constants were calculated using the canonical variational transition state theory following the procedure proposed by da Silva and Bozzelli [38]. Specifically, the positions of the optimal transition state vary with temperature.

3.2. Brief description of the kinetic model

In this work, the simulations of pyrolysis, oxidation in JSR, and ignition delay times were performed using Chemkin-Pro software [39]. The present model consisting of 160 species and 1038 reactions was mainly developed from our reported *n*-butanol model [3]. The *n*-butanol model [3] was developed from USC Mech II model [40] and three butene isomers model [18]. Most thermodynamic and transport data were taken from the USC Mech II model [40], the database reported by Goos et al. [41], and previous butanol models [8,16]. The reaction mechanism, thermodynamic and transport data are available in the Supplementary materials.

3.3. Sub-mechanism of 2-butanol

El-Nahas et al. [42] calculated the unimolecular decomposition reactions of 2-butanol using the ab initio methods (CBS-QB3 and CCSD(T)) and the density functional theory (DFT). The high-pressure limit rate constants of 2-butanol decomposition reactions were also calculated using the transition state theory (TST). Recently, Rosado-Reyes and Tsang [14] studied the thermal decomposition of 2-butanol using the single pulse shock tube technique at 1045–1221 K over pressure range of 1.5–6.0 bar. The rate constants of R1–R4 were obtained in their work.

$$sC_4H_9OH = 1\text{-}C_4H_8 + H_2O \tag{R1}$$

$$sC_4H_9OH = 2\text{-}C_4H_8(Z) + H_2O \tag{R2}$$

$$sC_4H_9OH = 2\text{-}C_4H_8(E) + H_2O \tag{R3}$$

$$sC_4H_9OH = C_2H_5 + CH_3CHOH \tag{R4}$$

$$sC_4H_9OH = CH_3 + CH_2CH(OH)CH_3 \tag{R5}$$

$$sC_4H_9OH = CH_3 + C_2H_5CHOH \tag{R6}$$

The calculated major decomposition pathways of 2-butanol at the QCISD(T) level are shown in Fig. 1. The rate constants of the channels R1–R6, which were calculated at 5, 30, 70, 200, 760, 7600 and 76,000 Torr in the temperature range of 800–2000 K, are listed in Table 2. The reaction rates of R1–R5 from the previous models developed by Moss et al. [8], Grana et al. [17], Sarathy et al. [16] and the present model are compared and illustrated in Fig. 2. The rate constants used in the models of Moss et al. [8], Sarathy et al. [16] and the present model have distinct differences. The calculated C—C scission reaction rate constants computed in the present work are very close to those measured by Rosado-Reyes and Tsang [14], as shown in Fig. 2. However, the calculated H_2O elimination reaction rate constants (R1)–(R3) are approximately 2–4 times higher than the measurements [14]. In order to get reliable rate constants for the primary reactions of 2-butanol decomposition, various electronic structure methods other than QCISD(T) including G3B3, CCSD and CBS-APNO have been employed to examine the accuracy of the computed energies. The uncertainties in the calculated energies by these three methods are within 1.1 kcal/mol compared with the QCISD(T) result, which ensures the reliability of the energies used for the rate constants calculations. Klippenstein et al. have suggested that the uncertainty of computed rate constants with variational transition state theory method could be a factor of 2 based on QCISD(T)/CBS potential energy surfaces [24]. In the present work, the reaction rate constants of R1–R3 are divided by a factor of 1.5–3 to accord with the measurements [14]. The pyrolysis simulation is very sensitive to these rate constants, which will be discussed in detail in the following section.

In the 2-butanol chemistry, H-abstraction reactions of 2-butanol by small radicals are important. In previous models, most rate constants for the sC_4H_9OH + H reactions were referred to analogous abstraction reactions such as ethanol + H or alkane + H. In Hansen et al. model [4], the rate constant of sC_4H_9OH + H reaction was calculated at the CBS-QB3 level, which was adopted by the present model.

The rate constants of sC_4H_9OH + OH have been estimated by Sarathy et al. [16] by combining available theoretical results, low-temperature experimental measurements, simple Evans-Polanyi type correlations and their understanding of alcohol and alkane combustion chemistry. Recently, Pang et al. [43] measured the OH time-histories in shock-heated mixtures of *tert*-butylhydroperoxide (TBHP) with 2-butanol excessively diluted in argon. The measured OH time-histories are well simulated using the Sarathy et al.'s mechanism [16]. Thus the rate constants of sC_4H_9OH + OH estimated by Sarathy et al. [16] are adopted by the present model.

The β-scission of sC_4H_8OH radicals were also calculated in this work. The pressure- and temperature-dependent rate constants are listed in Table 2 as well.

4. Results and discussion

4.1. 2-Butanol pyrolysis in flow reactor

The present pyrolysis experiments are simulated using the Plug Flow code in the Chemkin-Pro program [39]. The problem type is selected as "fix gas temperature". The temperature profiles along the flow tube are used as input parameters. The ending axial position is set to be 21.0 cm because the experimental data were sampled at this position. The pressure profiles are used in the simulation at 5 Torr. The volumetric flow rate is 1000 SCCM at 298.15 K. The residence times are calculated based on the input gas flow rates, temperature and pressure profiles. So the residence times are not used as input parameters in the simulation. The concentration of the inlet Ar is 97% and sC_4H_9OH is 3%. The simulated

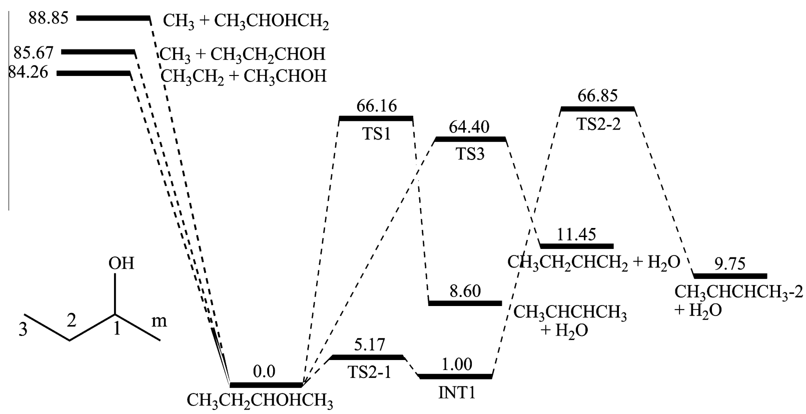

Fig. 1. The calculated major decomposition pathways of 2-butanol at the QCISD(T) level including the ZPE correction. (Energy unit: kcal/mol.)

results at 21.0 cm are then used to be compared with the experimental results.

The experimental and simulated mole fraction profiles of 2-butanol pyrolysis at various pressures are illustrated in Figs. 3–7. Generally, the present model predicts most of the detected species within the experimental uncertainty. The sensitivity analysis and rate of production (ROP) analysis at 5 Torr, 1450 K and 760 Torr, 1100 K were performed to obtain an insight for 2-butanol chemistry. Since the reaction pathways of 2-butanol pyrolysis are very similar in 5 Torr and 760 Torr, only the reaction network at 760 Torr is illustrated in Fig. 8.

The mole fraction profiles of 2-butanol are presented in Fig. 3. At 760 Torr, the decomposition temperature range is approximately 200 K (from 1000 K to 1200 K), while at 5 Torr the decomposition temperature range is approximately 300 K (from 1200 K to 1500 K). The ROP analysis shows that approximately 75% of 2-butanol decomposes through unimolecular pathways (R1)–(R6) at 5 Torr and this ratio decreases to 55% at 760 Torr. Sensitivity analysis of 2-butanol in Fig. 9 also shows that 2-butanol decomposition has great sensitivity to unimolecular reactions. The simulated mole fractions of 2-butanol using different 2-butanol unimolecular decomposition rate coefficients are illustrated in Fig. 10a. Changing the rate coefficients of R1–R6 artificially by a factor of 2 leads to approximately 30 K shift at 5 Torr, but only 10 K shift at 760 Torr. Because the unimolecular pathways are more important at low pressure, these pathways are mainly discussed at 5 Torr in this section. A comparison of the simulated results of 2-butanol decomposition at 760 Torr by the present model, Sarathy et al. model [16] and Grana et al. model [17] is shown in Fig. S3a in the Supplementary materials. All these models predict our measured mole fractions of 2-butanol very well while the variances are within 10 K. It also indicates the accuracy of the measured temperatures in this work.

About 40% of 2-butanol decomposes through H_2O elimination reactions producing 1-C_4H_8, 2-C_4H_8(Z) and 2-C_4H_8(E). Figure 3b–d show the experimental and simulated results of H_2O, 1-C_4H_8 and 2-C_4H_8. Since 2-C_4H_8(Z) and 2-C_4H_8(E) are indistinguishable in the experiment, the total mole fractions of 2-C_4H_8 are shown in Fig. 3d. The sensitivity analysis shows that the mole fractions of 1-C_4H_8 (Fig. 11) and 2-C_4H_8 (Fig. 12) are most sensitive to the rate constants of H_2O elimination reactions. Figure 10b also shows that 1-C_4H_8 formation is highly sensitive to k_{R1}. Significant deviations on the simulated 1-C_4H_8 concentrations were introduced by artificially enlarging or reducing k_{R1} by a factor of 2 both at 5 Torr and 760 Torr. Similar conclusions are obtained for 2-C_4H_8 concentrations when we utilized different k_{R2} and k_{R3} in the simulation. It indicates that the pressure-dependent pyrolysis experiment in the flow tube reactor is sensitive to the unimolecular reactions of the studied fuels; consequently it is suitable to validate the sub-mechanism of the fuel decomposition and primary products. The rate constant of R1 in Sarathy et al. model [16] is smaller than that used in the present model, which leads to the underprediction of 1-C_4H_8 concentration, as shown in Fig. S3b in the Supplementary materials. On the other hand, k_{R1} is overestimated in Grana et al. model [17], as a result, 1-C_4H_8 concentration is over-predicted by Grana et al. model [17].

Further decomposition of 1-C_4H_8 and 2-C_4H_8 leads to the formation of aC_3H_5, C_3H_6 and two C_4H_7 isomers ($CH_2{=}CH{-}CH^{*}CH_3$ and $CH_2{=}CH{-}CH_2{-}CH_2^{*}$). Figure 6 shows the experimental and simulated results of C3-species. The C3-speices are mainly produced by the reaction sequence of 1-C_4H_8/2-C_4H_8 → C_3H_6 → aC_3H_5 → aC_3H_4 → pC_3H_4. These pathways have been discussed in detail in pyrolysis of three butene isomers [18] and of *n*-butanol [3], and will not be discussed in this paper.

C1—C2 bond is the weakest bond in 2-butanol (see Fig. 1 for the atom label definitions). 33% of 2-butanol is consumed through this C—C bond scission (R4) producing C_2H_5 and sC_2H_4OH radicals ($CH_3CH^{*}OH$). Further destruction of C_2H_5 and sC_2H_4OH radicals leads to the formation of large amount of H atom, which then attacks 2-butanol and accelerates its decomposition. Thus, as shown in Fig. 9, R4 has the greatest negative sensitivity coefficient regard to the decomposition of 2-butanol. The other destruction of sC_2H_4OH radical is dominated by losing an H atom to produce C_2H_3OH (ethenol, R7, 30%) and CH_3CHO (R8, 65%). C_2H_3OH and CH_3CHO were detected and distinguished from each other in the experiment. Figure 5a and b show the experimental and simulated mole fractions of C_2H_3OH and CH_3CHO. A comparison of the simulation results at 760 Torr by using the present model, Sarathy et al. model [16] and Grana et al. model [17] is shown in Fig. S3 in the Supplementary materials. The maximum mole fraction of C_2H_3OH predicted by Sarathy et al. model [16] is approximately three times larger than the experimental result. However, it is worth noting that their simulated results approach to experimental results as the pressure gets lower. The predicted maximum mole fraction by Sarathy et al. model [16] at 5 Torr is within the experimental uncertainty. The ROP analysis shows that the formation of C_2H_3OH is dominated by the reaction sequence of sC_4H_9OH → sC_2H_4OH → C_2H_3OH at low pressure both in the present model and Sarathy et al. model. However, the reaction sequence of sC_4H_9OH → sC_4H_8OHm → C_2H_3OH plays an important role in the formation of C_2H_3OH at high pressure. Thus the over-predicted C_2H_3OH by Sarathy et al. model is possibly due to the over-estimation of the sC_4H_9OH → sC_4H_8OHm → C_2H_3OH sequence. Enol-ketone reactions were not included in Grana et al. model [17], thus only the mole fraction profile of CH_3CHO was compared with other models. It can be seen that both the

Table 2
Calculation results and selected reactions in 2-butanol model. Units are s^{-1}, cm^3, and cal/mol.

	Selected reactions	*A*	*n*	*E*	Pressure (Torr)	Ref.
Reactions of sC_4H_9OH						
1	$sC_4H_9OH = 1\text{-}C_4H_8 + H_2O$	2.27E+91	−22.98	105,501	5	This work[a]
		1.29E+83	−20.39	102,922	30	This work
		4.77E+77	−18.74	100,603	70	This work
		4.87E+69	−16.35	96,787	200	This work
		2.32E+58	−13.00	90,956	760	This work
		5.27E+37	−6.96	79,473	7600	This work
		7.77E+20	−2.08	69,611	76,000	This work
2	$sC_4H_9OH = 2\text{-}C_4H_8(E)+H_2O$	2.96E+92	−23.41	107,112	5	This work[b]
		2.15E+84	−20.84	104,699	30	This work
		8.65E+78	−19.20	102,449	70	This work
		7.91E+70	−16.79	98,662	200	This work
		1.72E+59	−13.33	92,661	760	This work
		1.05E+38	−7.12	80,924	7600	This work
		2.50E+20	−2.01	70,628	76,000	This work
3	$sC_4H_9OH = 2\text{-}C_4H_8(Z)+H_2O$	6.18E+93	−23.72	108,171	5	This work[b]
		8.94E+85	−21.23	105,971	30	This work
		2.80E+80	−19.55	103,653	70	This work
		4.41E+72	−17.21	100,027	200	This work
		1.22E+61	−13.78	94,096	760	This work
		1.39E+40	−7.65	82,518	7600	This work
		8.79E+22	−2.66	72,475	76,000	This work
4	$sC_4H_9OH = C_2H_5 + CH_3CHOH$	1.79E+79	−19.00	108,004	5	This work
		6.51E+81	−19.44	112,455	30	This work
		8.51E+80	−19.06	113,240	70	This work
		1.12E+78	−18.09	112,995	200	This work
		5.01E+71	−16.11	110,742	760	This work
		8.61E+55	−11.41	102,984	7600	This work
		2.18E+40	−6.85	94,257	76,000	This work
5	$sC_4H_9OH = CH_3 + C_2H_5CHOH$	3.82E+101	−25.84	122,606	5	This work
		6.07E+99	−24.97	124,718	30	This work
		1.16E+97	−24.04	124,540	70	This work
		7.31E+91	−22.38	123,085	200	This work
		3.71E+82	−19.54	119,288	760	This work
		1.04E+62	−13.43	108,980	7600	This work
		6.55E+41	−7.52	97,650	76,000	This work
6	$sC_4H_9OH = CH_3 + CH_2CH(OH)CH_3$	5.79E+110	−28.52	130,548	5	This work
		6.73E+105	−26.75	130,556	30	This work
		7.45E+101	−25.46	129,604	70	This work
		2.19E+95	−23.41	127,337	200	This work
		3.02E+84	−20.11	122,636	760	This work
		1.77E+61	−13.21	110,822	7600	This work
		3.24E+38	−6.57	98,022	76,000	This work
7	$sC_4H_9OH + H = sC_4H_8OH\text{-}1 + H_2$	4.14E+05	2.34	2680		[4]
8	$sC_4H_9OH + H = sC_4H_8OH\text{-}2 + H_2$	8.65E+05	2.30	4680		[4]
9	$sC_4H_9OH + H = sC_4H_8OH\text{-}3 + H_2$	1.68E+06	2.21	9580		[4]
10	$sC_4H_9OH + H = sC_4H_8OHm + H_2$	5.68E+06	2.21	7500		[4]
11	$sC_4H_9OH + H = sC_4H_9O + H_2$	5.01E+04	2.64	7150		[4]
12	$sC_4H_9OH + OH = sC_4H_8OH\text{-}1 + H_2O$	1.81E+03	2.89	−2611		[16]
13	$sC_4H_9OH + OH = sC_4H_8OH\text{-}2 + H_2O$	1.54E+00	3.70	−3736		[16]
14	$sC_4H_9OH + OH = sC_4H_8OH\text{-}3 + H_2O$	5.17E+06	1.81	12		[16]
15	$sC_4H_9OH + OH = sC_4H_8OHm + H_2O$	2.31E+00	3.70	−2946		[16]
16	$sC_4H_9OH + OH = sC_4H_9O + H_2O$	5.88E+02	2.82	−585		[16]
17	$sC_4H_9OH + CH_3 = sC_4H_8OH\text{-}1 + CH_4$	3.89E−01	3.53	4010		[4]
18	$sC_4H_9OH + CH_3 = sC_4H_8OH\text{-}2 + CH_4$	1.21E+01	3.43	9580		[4]
19	$sC_4H_9OH + CH_3 = sC_4H_8OH\text{-}3 + CH_4$	1.59E+00	3.62	13,430		[4]
20	$sC_4H_9OH + CH_3 = sC_4H_8OHm + CH_4$	1.42E+00	3.60	11,050		[4]
21	$sC_4H_9OH + CH_3 = sC_4H_9O + CH_4$	1.53E−02	3.92	7890		[4]
22	$sC_4H_9OH + HO_2 = sC_4H_8OH\text{-}1 + H_2O_2$	1.25E−05	5.26	7468		[16]
23	$sC_4H_9OH + HO_2 = sC_4H_8OH\text{-}2 + H_2O_2$	7.51E−03	4.52	14,710		[16]
24	$sC_4H_9OH + HO_2 = sC_4H_8OH\text{-}3 + H_2O_2$	4.14E−04	4.76	15,050		[16]
25	$sC_4H_9OH + HO_2 = sC_4H_8OHm + H_2O_2$	1.13E−02	4.52	13,850		[16]
26	$sC_4H_9OH + HO_2 = sC_4H_9O + H_2O_2$	6.47E−07	5.30	10,530		[16]
Reactions of sC_4H_8OH isomers						
27	$sC_4H_8OHm = C_4H_7OH1\text{-}2 + H$	1.78E+05	−0.21	13,413	5	This work
		1.35E+09	−1.02	15,882	30	This work
		1.01E+11	−1.41	17,230	70	This work
		4.68E+10	−1.16	16,862	200	This work
		7.24E+13	−1.84	19,531	760	This work
		1.96E+27	−5.23	31,053	7600	This work

(*continued on next page*)

Table 2 (*continued*)

	Selected reactions	A	n	E	Pressure (Torr)	Ref.
		1.68E+41	−8.73	44,200	76,000	This work
28	$sC_4H_8OHm = 1\text{-}C_4H_8 + OH$	4.20E+10	−1.02	12,649	5	This work
		8.15E+13	−1.74	15,115	30	This work
		4.23E+14	−1.84	15,818	70	This work
		4.58E+16	−2.29	17,550	200	This work
		4.48E+16	−2.13	17,776	760	This work
		4.40E+22	−3.53	23,562	7600	This work
		1.52E+39	−7.85	38,483	76,000	This work
29	$sC_4H_8OHm = C_2H_3OH + C_2H_5$	1.72E+09	−0.75	12,860	5	This work
		1.22E+10	−0.76	13,295	30	This work
		2.46E+11	−1.01	14,325	70	This work
		1.85E+13	−1.40	15,886	200	This work
		2.71E+19	−2.94	21,096	760	This work
		7.42E+26	−4.71	28,027	7600	This work
		3.75E+40	−8.22	40,680	76,000	This work
30	$sC_4H_8OH\text{-}1 = C_4H_7OH1\text{-}2 + H$	1.61E+21	−5.27	22,642	5	This work
		5.49E+25	−6.16	25,611	30	This work
		1.36E+28	−6.66	27,397	70	This work
		3.02E+31	−7.39	30,059	200	This work
		1.90E+36	−8.47	34,118	760	This work
		1.64E+53	−12.76	48,389	7600	This work
		5.84E+60	−14.41	56,993	76,000	This work
31	$sC_4H_8OH\text{-}1 = C_4H_7OH2\text{-}2 + H$	1.61E+21	−5.27	22,642	5	This work
		5.49E+25	−6.16	25,611	30	This work
		1.36E+28	−6.66	27,397	70	This work
		3.02E+31	−7.39	30,059	200	This work
		1.90E+36	−8.47	34,118	760	This work
		1.64E+53	−12.76	48,389	7600	This work
		5.84E+60	−14.41	56,993	76,000	This work
32	$sC_4H_8OH\text{-}1 = iC_3H_5OH + CH_3$	9.39E+12	−1.65	15,539	5	This work
		4.82E+15	−2.19	17,655	30	This work
		1.96E+17	−2.54	18,970	70	This work
		1.53E+16	−2.10	18,143	200	This work
		1.83E+24	−4.19	24,738	760	This work
		1.14E+30	−5.50	30,487	7600	This work
		8.63E+47	−10.20	45,737	76,000	This work
33	$sC_4H_8OH\text{-}1 = C_2H_5COCH_3 + H$	3.39E+12	−1.75	15,424	5	This work
		5.98E+13	−1.75	17,648	30	This work
		4.64E+16	−2.59	18,814	70	This work
		6.58E+18	−3.06	20,676	200	This work
		1.04E+22	−3.79	23,552	760	This work
		1.21E+29	−5.46	30,194	7600	This work
		1.00E+42	−8.74	42,339	76,000	This work
34	$sC_4H_8OH\text{-}2 = 2\text{-}C_4H_8 + OH$	3.85E+18	−3.36	14,184	5	This work
		2.64E+21	−3.94	16,419	30	This work
		1.04E+23	−4.28	17,747	70	This work
		2.11E+25	−4.80	19,737	200	This work
		7.99E+28	−5.64	22,908	760	This work
		2.82E+40	−8.58	33,125	7600	This work
		1.44E+43	−9.04	37,585	76,000	This work
35	$sC_4H_8OH\text{-}2 = CH_3 + CH_3CHCHOH$	6.10E+12	−1.59	15,922	5	This work
		3.24E+19	−3.24	21,555	30	This work
		4.50E+21	−3.74	23,470	70	This work
		4.22E+20	−3.33	22,935	200	This work
		5.53E+24	−4.30	26,888	760	This work
		1.56E+44	−9.42	43,980	7600	This work
		4.79E+43	−8.94	47,090	76,000	This work
36	$sC_4H_8OH\text{-}2 = C_4H_7OH2\text{-}2 + H$	1.92E+11	−1.52	16,139	5	This work
		1.68E+18	−3.22	21,907	30	This work
		2.97E+20	−3.73	23,899	70	This work
		1.88E+19	−3.27	23,245	200	This work
		1.95E+28	−5.59	31,187	760	This work
		3.85E+43	−9.54	44,976	7600	This work
		2.20E+43	−9.12	48,452	76,000	This work
37	$sC_4H_8OH\text{-}2 = C_4H_7OH1\text{-}3 + H$	1.49E+12	−1.85	19,624	5	This work
		1.32E+17	−2.94	23,349	30	This work
		7.11E+19	−3.56	25,590	70	This work
		2.95E+23	−4.40	28,718	200	This work
		6.26E+28	−5.67	33,559	760	This work
		5.41E+46	−10.30	49,387	7600	This work
		1.07E+48	−10.24	54,407	76,000	This work

Table 2 (*continued*)

	Selected reactions	A	n	E	Pressure (Torr)	Ref.
38	sC_4H_8OH-3 = C_4H_7OH1-3 + H	1.18E−11	3.78	14,600	5	This work
		1.14E−01	1.32	16,953	30	This work
		1.01E+00	1.23	16,083	70	This work
		1.00E+08	−0.78	20,968	200	This work
		5.74E+09	−0.98	21,387	760	This work
		7.78E+19	−3.31	29,219	7600	This work
		8.91E+31	−6.17	40,558	76,000	This work
39	sC_4H_8OH-3 = C_2H_4 + sC_2H_4OH	4.16E+08	−0.42	10,505	5	This work
		6.82E+10	−0.83	12,130	30	This work
		1.24E+12	−1.09	13,120	70	This work
		6.62E+13	−1.44	14,556	200	This work
		3.96E+16	−2.06	16,962	760	This work
		5.80E+22	−3.50	22,700	7600	This work
		3.85E+33	−6.22	33,033	76,000	This work
Reactions of sC_2H_4OH						
40	sC_2H_4OH = CH_3CHO + H	9.12E+22	−5.05	20,778	5	This work
		6.37E+26	−5.86	23,807	30	This work
		7.52E+28	−6.31	25,520	70	This work
		7.58E+31	−6.99	28,071	200	This work
		2.38E+36	−8.04	32,008	760	This work
		3.05E+51	−11.89	44,961	7600	This work
		2.02E+57	−13.13	52,127	76,000	This work
41	sC_2H_4OH = C_2H_3OH + H	1.61E+21	−5.27	22,642	5	This work
		5.49E+25	−6.16	25,611	30	This work
		1.36E+28	−6.66	27,397	70	This work
		3.02E+31	−7.39	30,059	200	This work
		1.90E+36	−8.47	34,118	760	This work
		1.64E+53	−12.76	48,389	7600	This work
		5.84E+60	−14.41	56,993	76,000	This work
42	sC_2H_4OH = C_2H_5O	5.48E+45	−11.63	44,328	0.76	[16]
		2.54E+49	−12.37	46,445	7.6	[16]
		1.65E+54	−13.40	50,330	76	[16]
		1.81E+55	−13.31	53,132	760	[16]
		4.58E+49	−11.32	52,714	7600	[16]
		4.11E+46	−10.33	51,834	15,200	[16]
		6.68E+41	−8.83	50,202	38,000	[16]
		6.54E+37	−7.58	48,697	76,000	[16]
43	sC_2H_4OH = C_2H_4OH	2.65E+36	−8.86	51,019	0.76	[16]
		3.56E+37	−8.89	51,114	7.6	[16]
		4.14E+39	−9.19	51,912	76	[16]
		5.82E+44	−10.34	55,296	760	[16]
		4.26E+48	−11.06	59,458	7600	[16]
		8.84E+47	−10.74	59,901	15,200	[16]
		2.23E+45	−9.84	59,604	38,000	[16]
		1.70E+42	−8.83	58,737	76,000	[16]
44	sC_2H_4OH + O_2 = CH_3CHO + HO_2	5.26E+17	−1.64	838	7.6	[45]
		5.26E+17	−1.64	838	76	[45]
		5.28E+17	−1.64	839	760	[45]
		1.54E+18	−1.77	1120	7600	[45]
		3.78E+20	−2.43	3090	76,000	[45]
45	sC_2H_4OH + O_2 = C_2H_3OH + HO_2	5.12E+02	2.50	−414	7.6	[45]
		5.33E+02	2.50	−402	76	[45]
		7.62E+02	2.45	−296	760	[45]
		8.92E+03	2.15	470	7600	[45]
		4.38E+05	1.70	2330	76,000	[45]

[a] The reaction rates are divided by a factor of 3 to accord with that measured by Rosado-Reyes and Tsang [14].
[b] The reaction rates are divided by a factor of 1.5 to accord with that measured by Rosado-Reyes and Tsang [14].

present model and Grana et al. model [17] can predict the trends of CH_3CHO at 760 Torr within the experimental uncertainties. The ROP analysis shows that the tautomerization reaction (R9) is the dominant pathway of ethenol consumption. The further decomposition of CH_3CHO is dominated by the reaction sequence of $CH_3CHO \rightarrow CH_3CO \rightarrow CO$, which contributes to about 90% of CO formation.

$$sC_2H_4OH = C_2H_3OH + H \tag{R7}$$

$$sC_2H_4OH = CH_3CHO + H \tag{R8}$$

$$C_2H_3OH = CH_3CHO \tag{R9}$$

C1—Cm scission (R5, 2% of 2-butanol consumption) and C2—C3 scission (R6, 1%) are secondary 2-butanol unimolecular decomposition pathways when compared to the C1—C2 scission (R4, 33%). C_2H_5CHOH and $CH_2CH(OH)CH_3$ are the products of R5 and R6,

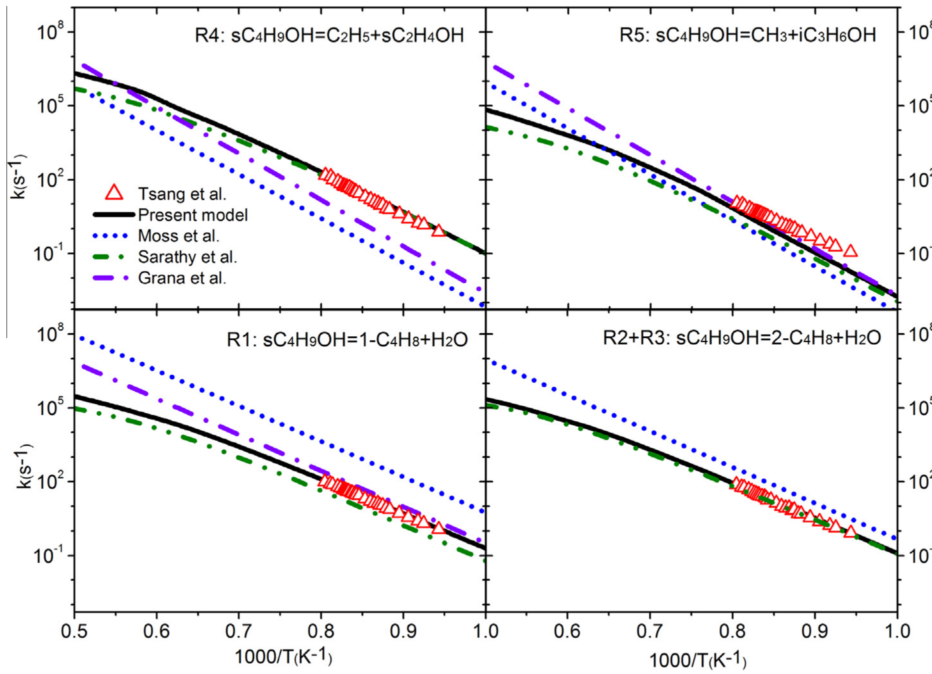

Fig. 2. Reaction rate constants of R1–R5 at 760 Torr in the present model, Moss et al. model [8], Sarathy et al. model [16], Grana et al. model [17] and Rosado-Reyes and Tsang's measurement [14].

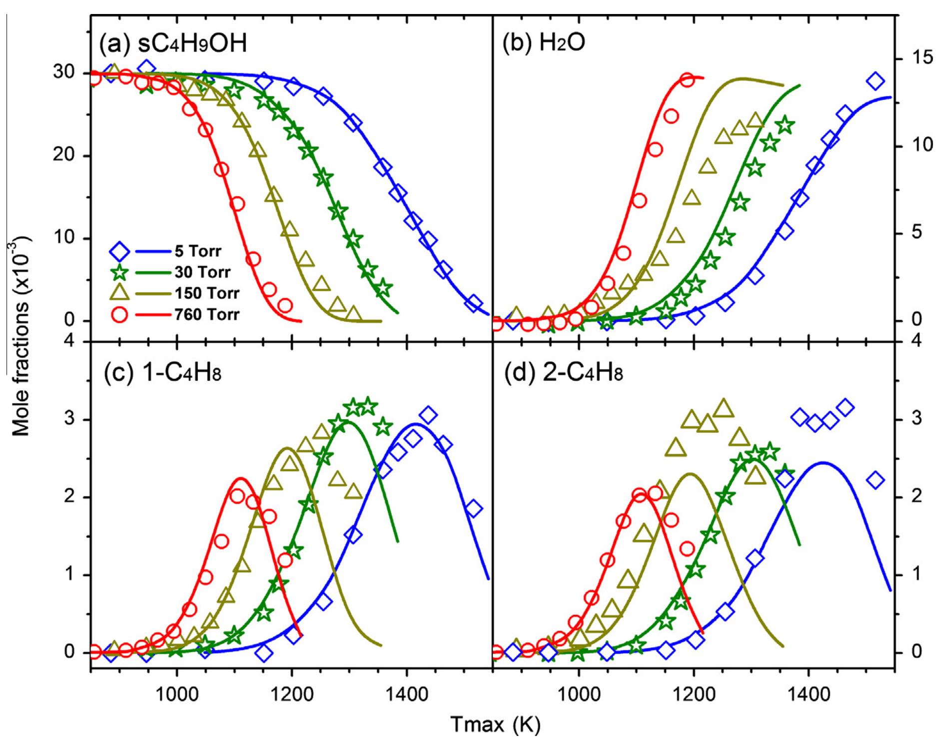

Fig. 3. The mole fraction profiles measured in this work (symbols) and simulated results (lines) of sC_4H_9OH, H_2O, 1-C_4H_8 and 2-C_4H_8 from 2-butanol pyrolysis at various pressures.

respectively, the decomposition of which produces C_2H_3OH. These pathways contribute approximately 15% to C_2H_3OH formation.

About 45% of 2-butanol decomposes through H-abstraction reactions by small radical attack at 760 Torr, while this ratio is only 24% at 5 Torr. So the H-abstraction and its follow-up pathways are discussed merely at 760 Torr in this section. As shown in Figs. 3–5, the mole fractions of intermediate species from unimolecular decomposition pathways are higher at low pressure than those at high pressure, such as 1-C_4H_8 and C_2H_3OH. However, the mole fractions of intermediate species from H-abstraction pathways (e.g., CH_3COCH_3 and $C_2H_5COCH_3$) show the contrary trend.

sC_4H_8OH-1 is the most dominant product of the H-abstraction reactions. Approximately 95% of sC_4H_8OH-1 decomposes by β-C—C-scission reaction producing iC_3H_5OH (CH_2=CH(OH)—CH_3,

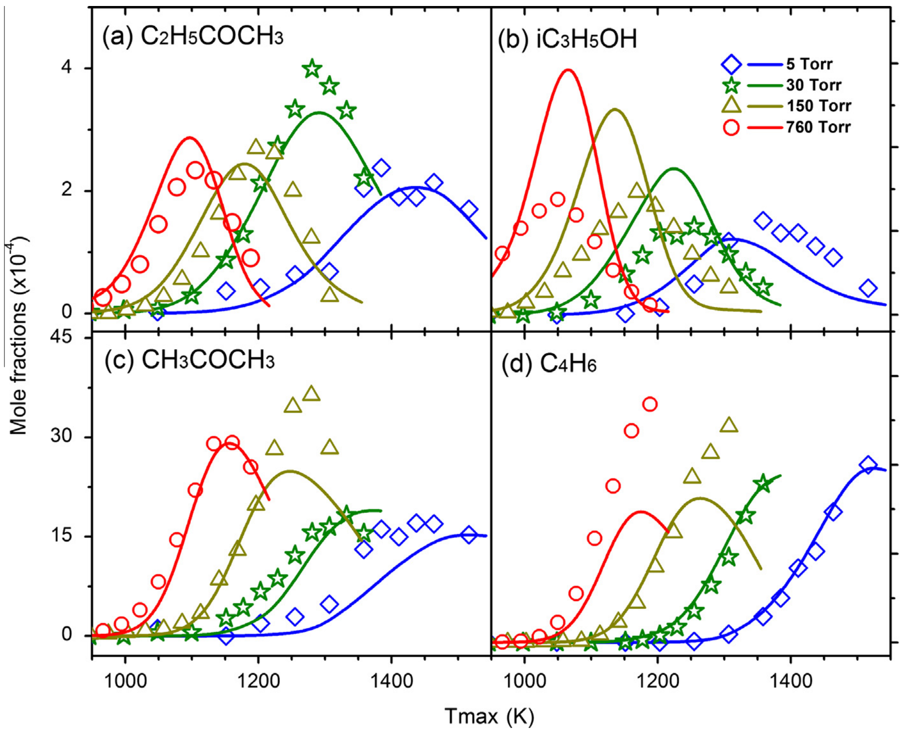

Fig. 4. The mole fraction profiles measured in this work (symbols) and simulated results (lines) of $C_2H_5COCH_3$, iC_3H_5OH, CH_3COCH_3 and C_4H_6 from 2-butanol pyrolysis at various pressures.

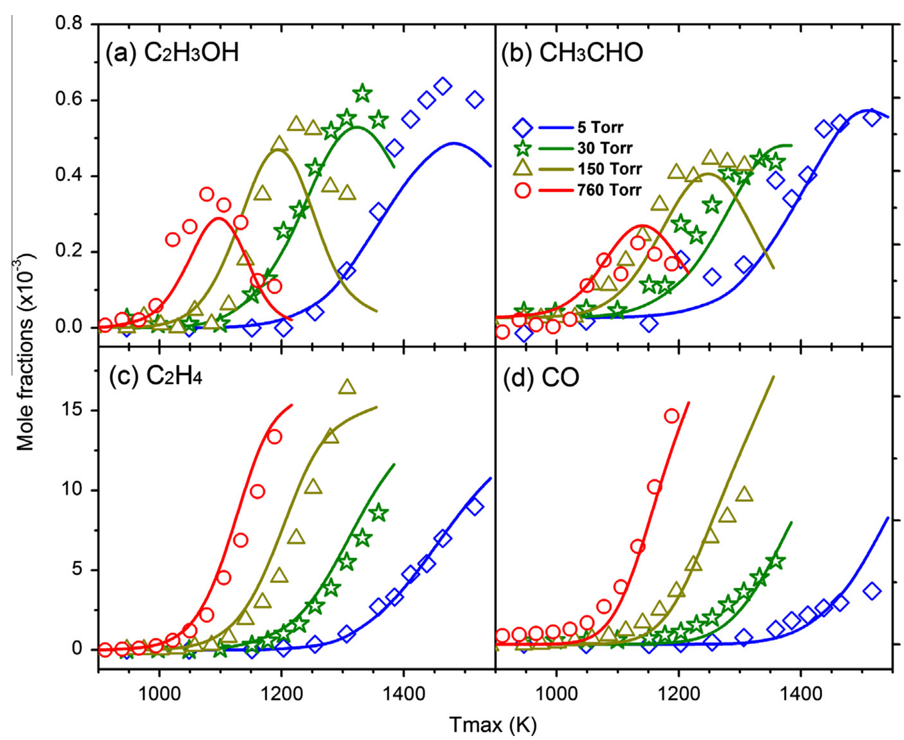

Fig. 5. The mole fraction profiles measured in this work (symbols) and simulated results (lines) of C_2H_3OH, CH_3CHO, C_2H_4 and CO from 2-butanol pyrolysis at various pressures.

R10), which is also the dominant pathway of iC_3H_5OH formation. A small amount of sC_4H_8OH-1 decomposes by β-C—H-scission producing $C_2H_5COCH_3$ (R11). Both iC_3H_5OH and $C_2H_5COCH_3$ were detected in the experiment, as shown in Fig. 4. The mole fractions of iC_3H_5OH are approximately two times larger than those of $C_2H_5COCH_3$. Most of iC_3H_5OH is consumed through the tautomerization reaction producing more stable species CH_3COCH_3 (R12). CH_3COCH_3 is distinguished from iC_3H_5OH in the experiment, and the

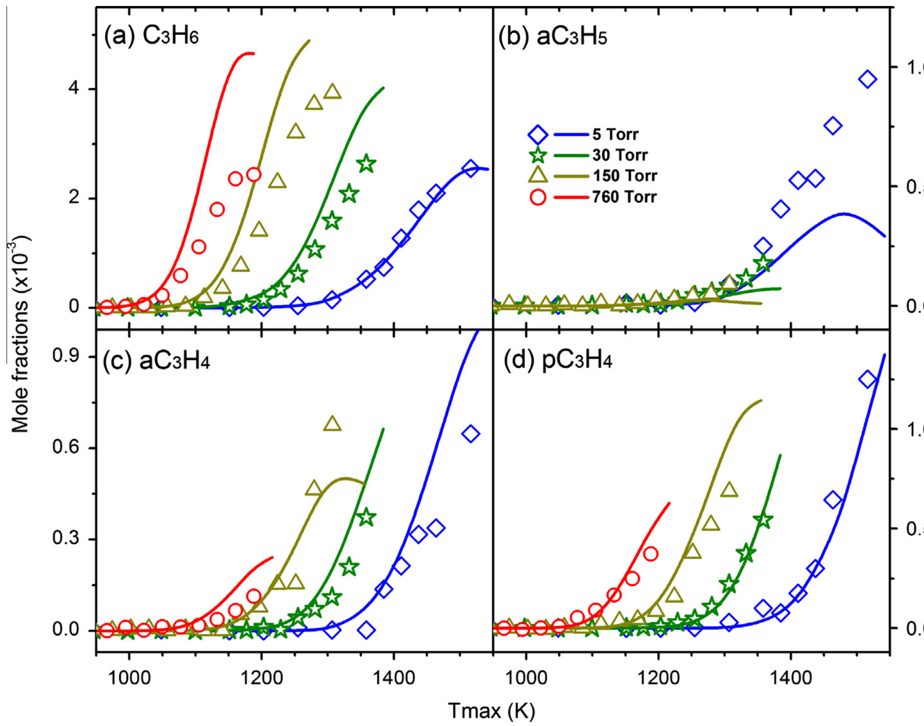

Fig. 6. The mole fraction profiles measured in this work (symbols) and simulated results (lines) of C_3H_6, aC_3H_5, aC_3H_4 and pC_3H_4 from 2-butanol pyrolysis at various pressures.

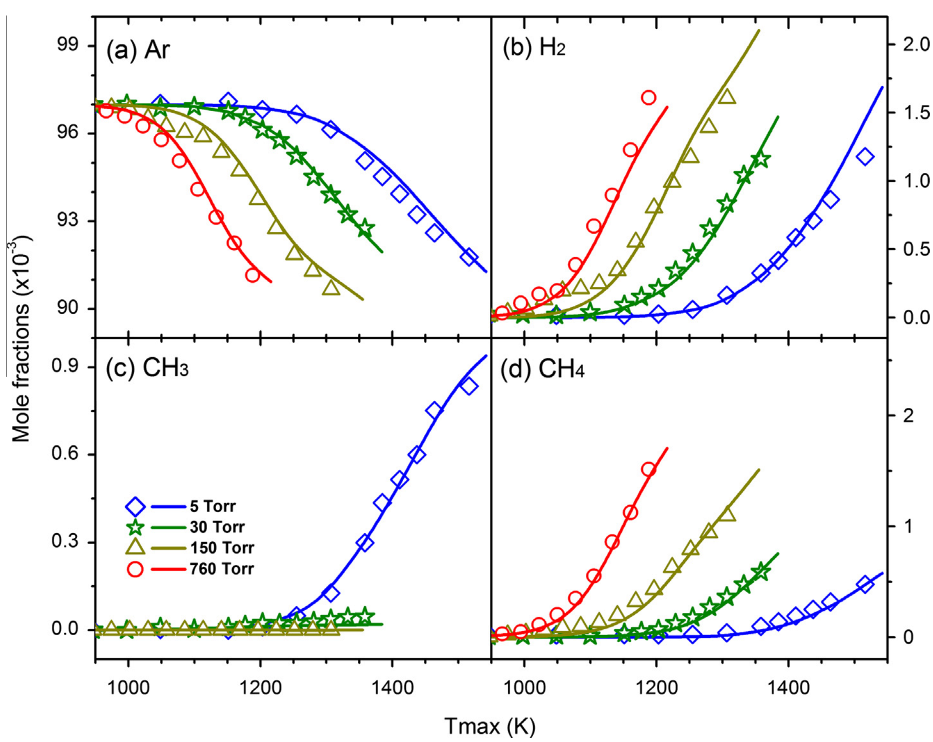

Fig. 7. The mole fraction profiles measured in this work (symbols) and simulated results (lines) of Ar, H_2, CH_3 and CH_4 from 2-butanol pyrolysis at various pressures.

results are shown in Fig. 4 as well. The further decomposition of CH_3COCH_3 is dominated by the reaction sequence of $CH_3COCH_3 \rightarrow CH_3COCH_2 \rightarrow CH_2CO \rightarrow CO$.

$$sC_4H_8OH\text{-}1 = iC_3H_5OH + CH_3 \quad (R10)$$

$$sC_4H_8OH\text{-}1 = C_2H_5COCH_3 + CH_3 \quad (R11)$$

$$iC_3H_5OH = CH_3COCH_3 \quad (R12)$$

The β-scission of sC_4H_8OH-2 leads to the formation of 2-C_4H_8 (R13) and $CH_3CHCHOH$ (R14). Similar to the C_2H_3OH decomposition, most of $CH_3CHCHOH$ is consumed by the tautomerization reaction producing C_2H_5CHO (R15). However, the mole fractions of $CH_3CHCHOH$ and C_2H_5CHO are approximately an order of magnitude smaller

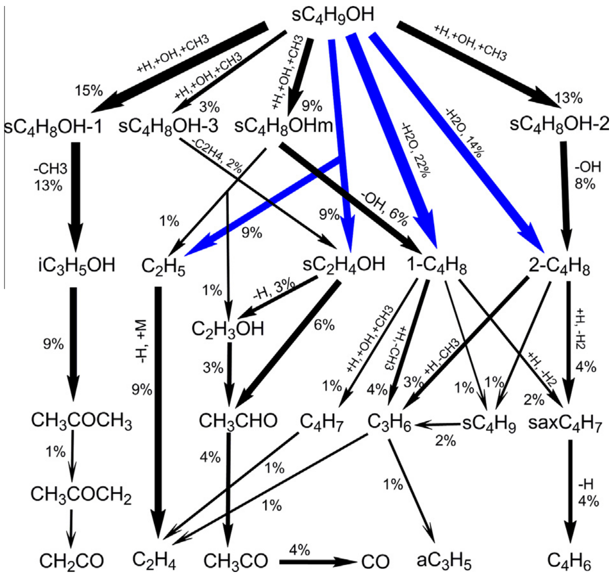

Fig. 8. Reaction network of sC_4H_9OH decomposition in pyrolysis at T_{max} = 1100 K and 760 Torr. Thickness of each arrow represents the carbon flux of corresponding reaction(s). The percentage in the figure means the carbon flux of the pathway divided by total carbon flux of sC_4H_9OH consumption. The blue arrows are the unimolecular pathways.

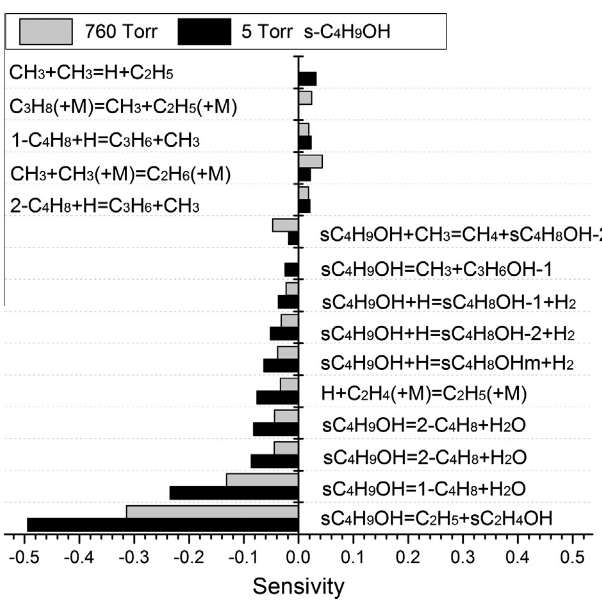

Fig. 9. Sensitivity analysis of sC_4H_9OH in pyrolysis at T_{max} = 1450 K under 5 Torr (gray) and T_{max} = 1100 K under 760 Torr (black). Only those reactions with sensitivities greater than 0.02 are listed.

than those of CH_3COCH_3 at all pressures. The destruction of sC_4H_8-OHm is also dominated by β-scission producing 1-C_4H_8 (R16) and C_2H_3OH (R17). R16 and R17 only contribute approximately 20% to the formation of 1-C_4H_8 and C_2H_3OH even at 760 Torr. Most of 1-C_4H_8 and C_2H_3OH are produced by reaction sequences of sC_4H_9-OH → 1-C_4H_8 and sC_4H_9OH → sC_2H_4OH → C_2H_3OH, respectively.

$$sC_4H_8OH\text{-}2 = 2\text{-}C_4H_8 + OH \quad (R13)$$

$$sC_4H_8OH\text{-}2 = CH_3CHCHOH + CH_3 \quad (R14)$$

$$CH_3CHCHOH = C_2H_5CHO \quad (R15)$$

$$sC_4H_8OHm = 1\text{-}C_4H_8 + OH \quad (R16)$$

$$sC_4H_8OHm = C_2H_3OH + C_2H_5 \quad (R17)$$

4.2. 2-Butanol pyrolysis in shock tube

The pyrolysis of 2-butanol behind reflected shock waves was investigated very recently by Stranic et al. [15]. The initial reflected shock temperatures and pressures for the experiments covered 1270–1640 K and 1.3–1.9 atm, and species time history measurements of OH, H_2O, C_2H_4, CH_4, and CO were acquired using UV and IR laser absorption method. The authors also simulated their experimental data using Sarathy et al. model [16], but the model cannot predict the species time history satisfactorily. Thus the authors modified some reactions in Sarathy et al. model according to the results of sensitivity analysis to get a relatively better prediction. However, there are still discrepancies between experimental and simulation results, which are beyond the experimental uncertainties.

The experimental results of Stranic et al. [15] were also validated by our present model. The simulations were conducted in Chemkin-Pro using adiabatic Constant-Volume (CV) model by solving energy equation, the simulated temperature time histories were then used to calculated species mole fraction time histories, as proposed by Stranic et al. [15,44]. Rate of production analysis and sensitivity analysis were also performed to assist further investigation of the pyrolysis behavior of 2-butanol. The analyses were all conducted at 1498 K and 1.76 atm since the main reaction pathways and sensitive reactions are almost identical at all experimental conditions.

Figure 13 presents the experimental and simulated results of OH and H_2O time history during 2-butanol pyrolysis behind shock waves. It can be seen that the present model can predict these species well within the experimental uncertainties. The mole fraction of OH reaches the peak value at approximately 2 μs and subsequently it begins to decrease until the equilibrium is reached. Figure 14a presents the sensitivity analysis of OH radical. The most sensitive reaction for OH production at early times is the unimolecular C1—C2 bond dissociation reaction (R4) that leads to C_2H_5 and sC_2H_4OH. A variation of the rate constant of R4 by a factor of 2 would lead to approximately 30% change of the peak value of OH production, which is shown in Fig. 15a. The OH mole fractions are also sensitive to the H abstraction reactions that lead to sC_4H_8-OH radicals. At early times, OH is mainly consumed by participating in the H-abstraction reaction of 2-butanol, which leads to sC_4H_8OH-1 radical. The sensitivity of this reaction for OH consumption becomes smaller as time increases. It should be noted that R4 becomes the most sensitive reaction for OH consumption as time increases, which is just opposite to its behavior at early times.

Both ROP and sensitivity analysis show that H_2O-elimination reactions (R1)–(R3) are the dominant reactions for H_2O production at all time scales, as shown in Fig. 14b. This is also the case in the flow reactor pyrolysis of 2-butanol, which has been discussed in the above section. A variation of the rate constants of R1–R3 leads to approximately 30% change of the concentration of H_2O, as shown in Fig. 15b. Thus, it indicates that the rate constants of R1–R3 are reasonable according to our flow reactor experiments and shock tube experiments performed by Rosado-Reyes and Tsang [14] and Stranic et al. [15].

The model predictions compared to experimental data for the formation of C_2H_4 and CH_4 behind shock waves are shown in Fig. 16. It can be seen that the present model can predict the time history of these two species within experimental uncertainties. The sensitivity analysis shows that R4 is the most important reaction

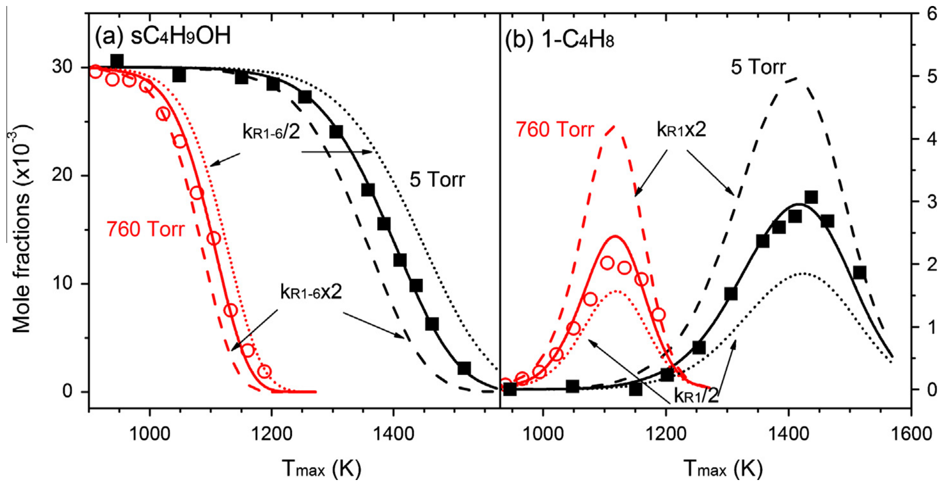

Fig. 10. (a) Experimental mole fraction profiles (symbols) and simulated results (lines) of sC_4H_9OH in pyrolysis at 5 and 760 Torr. The dash lines and dotted lines are the simulation results of sC_4H_9OH by using different rate coefficients of 2-butanol overall unimolecular decomposition (R1)–(R6). (b) Experimental mole fraction profiles (symbols) and simulated results (lines) of 1-butene (1-C_4H_8) in 2-butanol pyrolysis at 5 and 760 Torr. The dash lines and dotted lines are the simulation results of 1-C_4H_8 using different rate coefficients of sC_4H_9OH = 1-C_4H_8 + H_2O (R1). Solid lines are the simulated results from the present model.

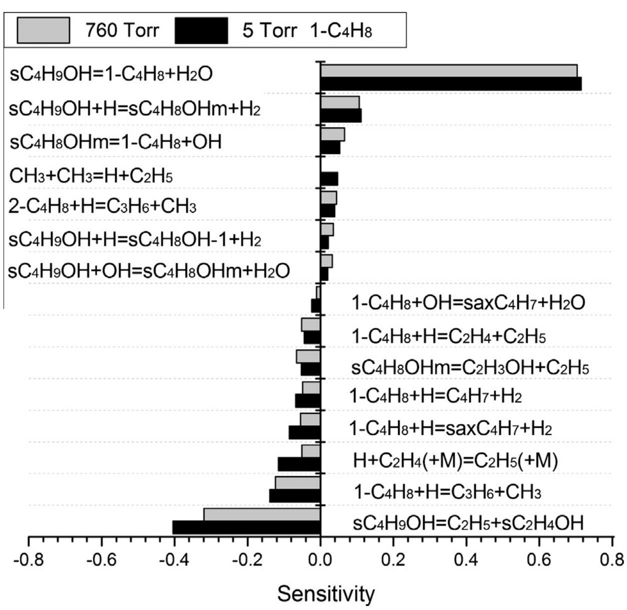

Fig. 11. Sensitivity analysis of 1-C_4H_8 in 2-butanol pyrolysis at T_{max} = 1450 K under 5 Torr (gray) and T_{max} = 1100 K under 760 Torr (black). Only those reactions with sensitivities greater than 0.02 are listed.

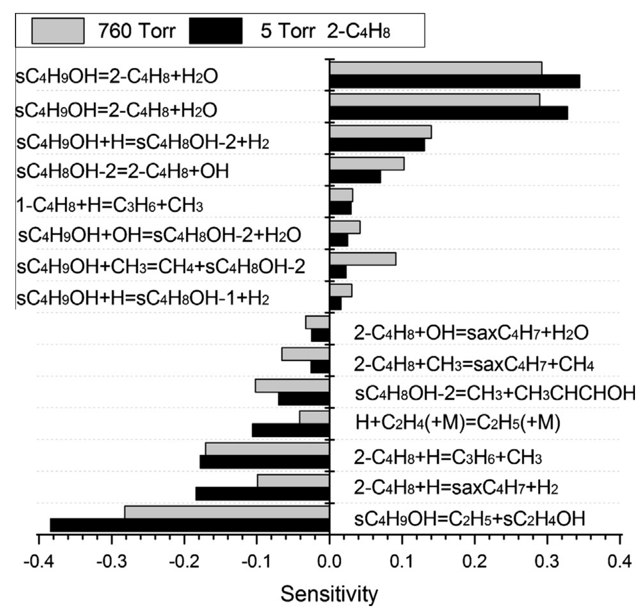

Fig. 12. Sensitivity analysis of 2-C_4H_8 in 2-butanol pyrolysis at T_{max} = 1450 K under 5 Torr (gray) and T_{max} = 1100 K under 760 Torr (black). Only those reactions with sensitivities greater than 0.02 are listed.

for C_2H_4 production, which is also the same case in the flow reactor pyrolysis. The most sensitive reaction for the production of CH_4 is also the unimolecular reaction (R5) of sC_4H_9OH, while ROP analysis shows that the H-abstraction reaction of CH_3CHO by CH_3 contributes almost half to CH_4 production, which indicates that the sequence of $sC_4H_9OH \rightarrow sC_2H_4OH \rightarrow CH_3CHO \rightarrow CH_4$ is the major pathway to produce CH_4. The H-abstraction of C_2H_4 by CH_3 is also an important route for the formation of CH_4, which contributes approximately 20% to CH_4 production.

The 2-butanol pyrolysis behind reflected shock waves in the pressure range of 1.2–2.2 atm over 1000–1500 K was also studied by Yasunaga et al. [10]. The concentrations of many pyrolysis species were measured. These data were also simulated by the present model and illustrated in Fig. S7 of the Supplementary materials.

4.3. Premixed 2-butanol flame

A rich laminar premixed 2-butanol flame at 30 Torr was studied by Yang et al. using the technique of SVUV-PIMS [12]. Many combustion intermediates including enols were identified by photoionization efficiency (PIE) spectra. Later, Oßwald et al. measured the mole fraction profiles of a rich premixed 2-butanol flame as a function of burner distance [13]. Both PI-MBMS and EI-MBMS techniques were used in their measurements. The present model is further validated against these measurements. The simulation was performed using the Laminar Premixed Flame Code in the Chemkin-Pro software. The experimental temperature profile was used in the simulation with a shift of 1.0 mm away from the burner surface to account for the perturbation induced by the quartz probe and the thermocouple.

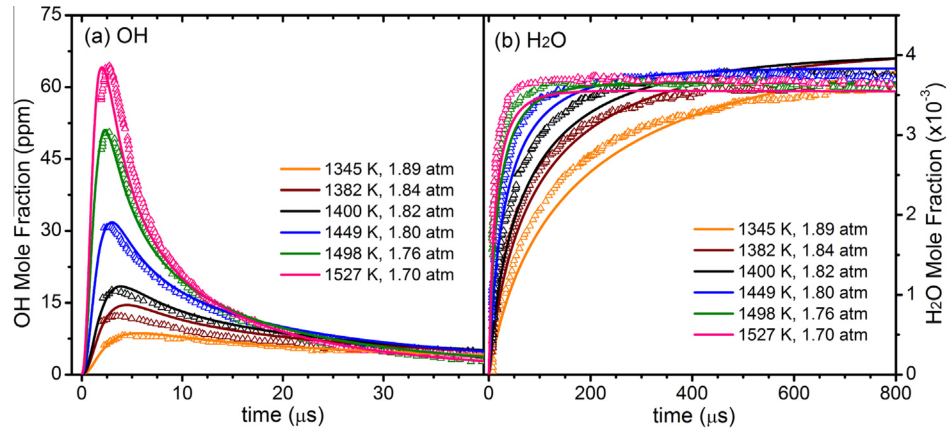

Fig. 13. Measured [15] (symbols) and simulated (solid lines) mole fraction profiles of OH and H_2O in the shock tube pyrolysis of 2-butanol. The uncertainty for OH and H_2O is approximately ±5%.

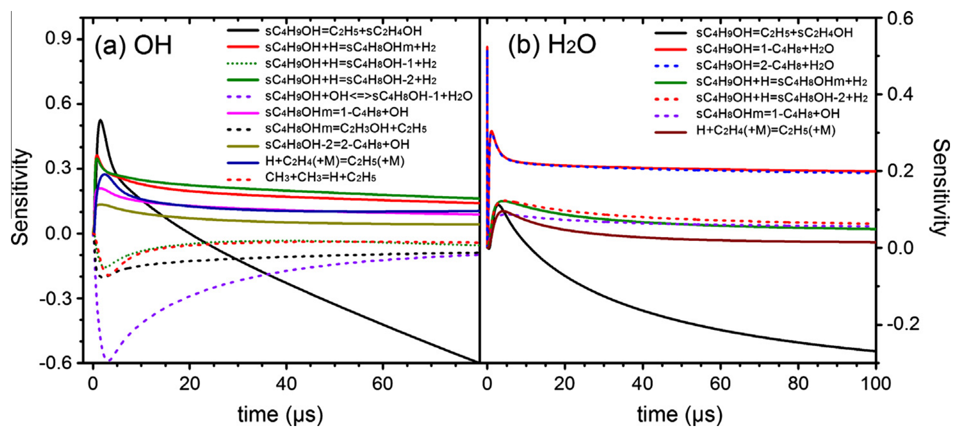

Fig. 14. Sensitivity analysis of 2-butanol shock tube pyrolysis. Initial conditions: T = 1498 K, P = 1.76 atm.

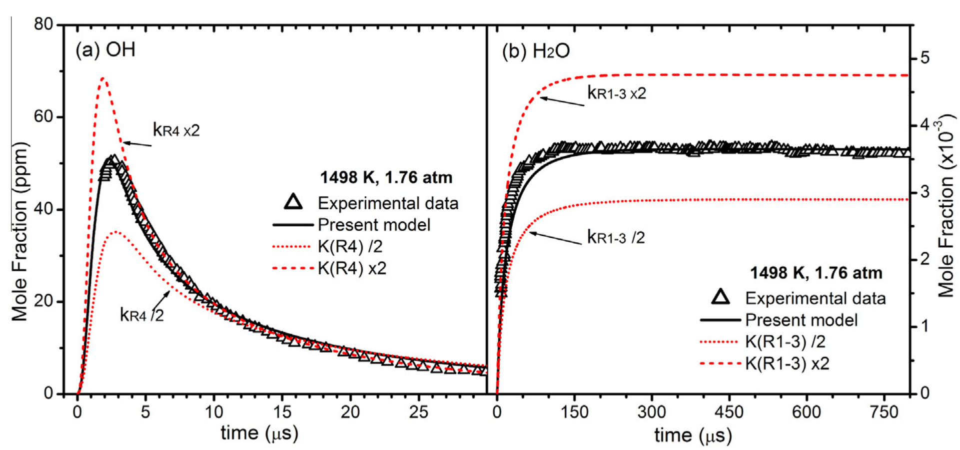

Fig. 15. (a) Experimental mole fraction profiles (symbols) and simulated results (lines) of OH in the shock tube pyrolysis at 1498 K and 1.76 atm. The dash lines and dotted lines are the simulation results of OH by using different rate coefficients of $sC_4H_9OH = C_2H_5 + CH_3CHOH$ (R4). (b) Experimental mole fraction profiles (symbols) and modeling results (lines) of H_2O at 1498 K and 1.76 atm. The dash lines and dotted lines are the simulation results of H_2O using different rate coefficients of H_2O elimination reactions (R1)–(R3). Solid lines are the simulated results from the present model.

Figure 17 shows the experimental and simulated results of major species in the rich 2-butanol flame. The present model can accurately predict the mole fractions of inert gas (Ar), reactants (sC_4H_9OH and O_2), and major products (H_2, H_2O, CO, and CO_2). The mole fraction profiles of selected C1—C4 flame intermediates are presented in Fig. 18. Generally, the present model can reproduce most of the experimental results quite well. The reaction network of the rich 2-butanol flame is illustrated in Fig. 19. It is

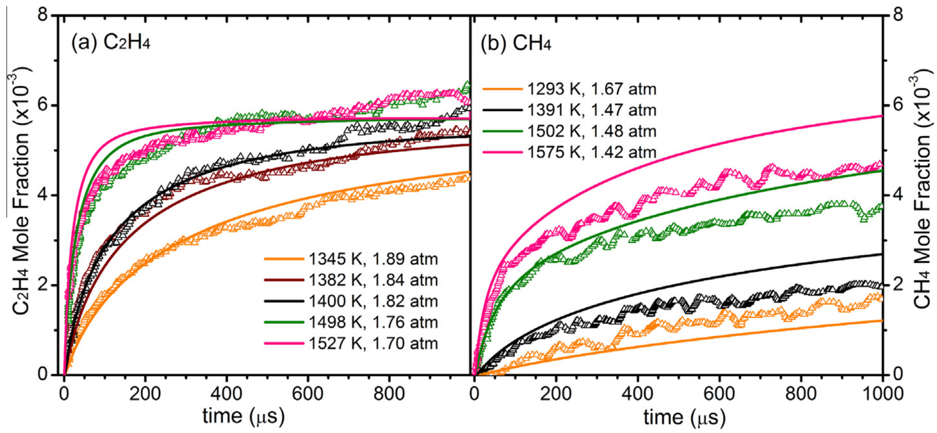

Fig. 16. Measured [15] (symbols) and simulated (solid lines) mole fraction profiles of C_2H_4 and CH_4 in the shock tube pyrolysis of sC_4H_9OH. The uncertainty for C_2H_4 and CH_4 is approximately ±10% and ±15%, respectively.

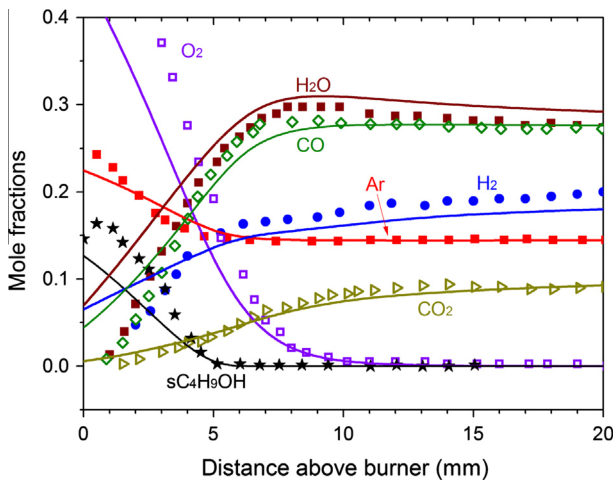

Fig. 17. Measured [13] (symbols) and simulated (solid lines) mole fraction profiles of major species in the premixed rich sC_4H_9OH flame at 30 Torr. The uncertainty of the experimental data is estimated to ±20%.

concluded that the reaction network in the premixed rich flame is quite similar to that in the pyrolysis. The following discussions will focus on the difference between pyrolysis and flame.

ROP analysis shows that 2-butanol is consumed by unimolecular reactions (35%) and H-abstraction reactions (65%) in the flame. The unimolecular reaction pathways are similar to those in pyrolysis and are not discussed in this section. In the 2-butanol flame, 37% of fuel decomposes through H-abstraction reactions by H atom, and the most preferred products are sC_4H_8OH-1 and sC_4H_8OHm, which is shown in Fig. 19. sC_4H_8OH-1 further decomposes mainly via two channels, which produce iC_3H_5OH and $C_2H_5COCH_3$, respectively. Figure 18a shows the mole fraction profiles of C3 and C4 oxygenated species. iC_3H_5OH and $C_2H_5COCH_3$ are the follow-up products of sC_4H_8OH-1 in both pyrolysis and flame of 2-butanol. Under the pyrolysis condition, the mole fractions of iC_3H_5OH are approximately two times larger than those of $C_2H_5COCH_3$ as mentioned above (see Fig. 4). However, in the 2-butanol flame experiment, the measured mole fraction of $C_2H_5COCH_3$ is approximately 6 times larger than those of iC_3H_5OH (Fig. 18). In the pyrolysis, sC_4H_8OH-1 is dominantly deconstructed by β-scission reactions producing iC_3H_5OH (82%) and $C_2H_5COCH_3$ (18%). However, except for β-scission reactions, sC_4H_8OH-1 is also consumed by reacting with O_2 followed by breaking the O-H bond (R18, 30% of sC_4H_8OH-1 consumption) in flames. As a result, in the flame, the formation channel of $C_2H_5COCH_3$ from sC_4H_8OH-1 is more competitive than that in pyrolysis. Approximately 52% of sC_4H_8OH-1 decomposes through the reaction sequence of $sC_4H_8OH\text{-}1 \rightarrow iC_3H_5OH \rightarrow CH_3COCH_3 \rightarrow CH_3COCH_2 \rightarrow CH_2CO$. This reaction sequence contributes more than 90% to the CH_3COCH_3 and CH_2CO formation. It is noticeable that the consumption of iC_3H_5OH is faster than that of $C_2H_5COCH_3$, thus the mole fraction of $C_2H_5COCH_3$ is much higher than that of iC_3H_5OH in the flame. We also compared the simulation results of $C_2H_5COCH_3$ and iC_3H_5OH by the present model and previous models [16,17]. The detailed information is provided in Fig. S4 of the Supplementary materials.

The production and decomposition pathways of CH_3COCH_3 are very similar in both pyrolysis and flame conditions. However, CH_3COCH_3 is well predicted in pyrolysis, but it is under-predicted by a factor of 3 in the flame. Furthermore, CH_3COCH_3 is also under-predicted by a factor of 4 in *tert*-butanol flame but well predicted in *tert*-butanol pyrolysis [19]. A possible reason is that some oxidation pathways are ignored in the present model, which may contribute to the decomposition of CH_3COCH_3 in the flame, but not exist in the pyrolysis. The other reason is that the experimental results of CH_3COCH_3 could be over-measured.

$$sC_4H_8OH\text{-}1 + O_2 = C_2H_5COCH_3 + HO_2 \quad \text{(R18)}$$

Approximately 60% of $1\text{-}C_4H_8$ and $2\text{-}C_4H_8$ are produced from the β-scission of sC_4H_8OH isomers. In Fig. 18, the total simulated mole fractions of $1\text{-}C_4H_8$ and $2\text{-}C_4H_8$ are compared to the measured C_4H_8. $1\text{-}C_4H_8$ and $2\text{-}C_4H_8$ are important precursors of C3 and C4 hydrocarbons.

C_2H_3OH and CH_3CHO were distinguished in the work of Oβwald et al. [13]. As presented in Fig. S4, the mole fractions of C_2H_3OH are over-predicted by both the present model and Sarathy et al. model [16]. The over-prediction of the reaction sequence of $sC_4H_9OH \rightarrow sC_2H_4OH/sC_4H_8OHm \rightarrow C_2H_3OH$ may lead to these results. However, it is worth noting that in the previous *n*-butanol kinetic studies [3,16], C_2H_3OH is also over-predicted in the flame after considering the tautomerization reaction by H, HO_2 and unimolecular reactions [3–5,16,45–47]. So the over-prediction of C_2H_3OH is possibly resulted from the absence of complex oxidation reactions or tautomerization reactions of other species.

4.4. 2-Butanol oxidation in jet-stirred reactor

The experiments on the oxidation of 2-butanol in a jet-stirred reactor at 10 atm over a wide range of equivalence ratios (0.5–4.0) have been performed by Togbé et al. [11]. Concentration profiles of reactants, stable intermediates and products were

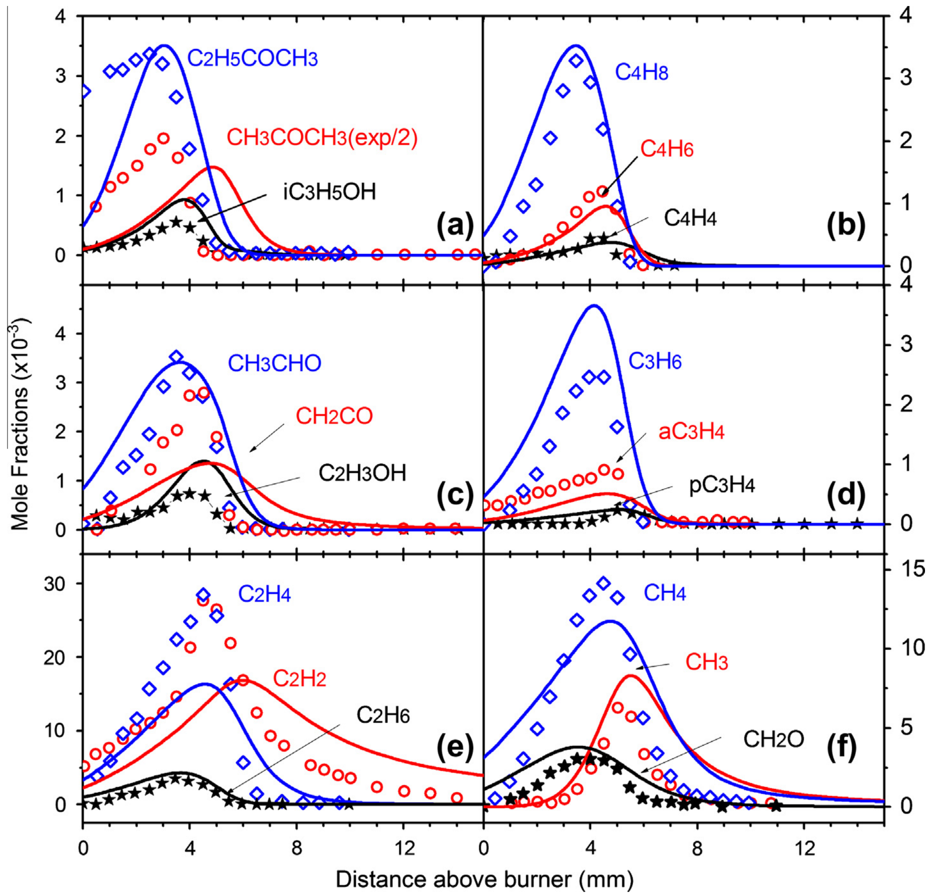

Fig. 18. Measured [13] (symbols) and simulated (solid lines) mole fraction profiles of C1–C4 intermediates in the premixed rich sC_4H_9OH flame at 30 Torr. The uncertainty of the experimental data is estimated to ±20%.

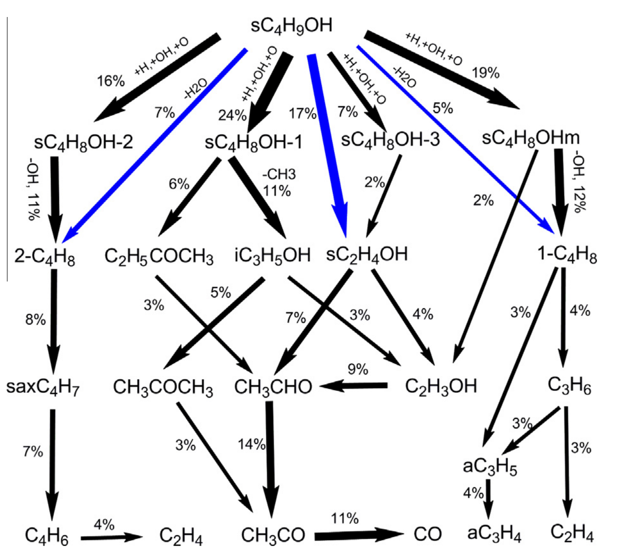

Fig. 19. Reaction network of the premixed rich sC_4H_9OH flame at 30 Torr. Thickness of each arrow represents the carbon flux of corresponding reaction(s). The percentage in the figure means the carbon flux of the pathway divided by total carbon flux of sC_4H_9OH consumption. The blue arrows are the unimolecular pathways. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

measured in the temperature range of 770–1250 K by the gas chromatography and infrared spectroscopy. These data were used for further validation of the present model. The residence time in the experiment is 0.7 s. The simulation was performed using the Perfectly Stirred Reactor code in Chemkin-Pro program with an end time of 100 s to obtain the steady-state mole fractions of oxidation species.

The experimental and simulated results of 2-butanol oxidation are presented in Figs. 20–22. The predicted mole fractions of most species are within a factor 2 of the experimental data. Some species have two peaks especially under $\phi = 4.0$ condition, such as 1-C_4H_8, H_2O, CO and C_2H_4. To analyze this interesting phenomenon, the ROP analysis was performed at 825 K and 1025 K, respectively. The ROP analysis at 825 K shows that more than 99% of 2-butanol decomposes through H-abstraction reactions by small radicals such as H (11%), OH (77%) and HO_2 (7%). sC_4H_8OH-1 radical is the most dominant product of the H-abstraction reactions (47% of 2-butanol consumption). About 63% of sC_4H_8OH-1 decomposes by reaction with O_2 producing $C_2H_5COCH_3$ and large amount of HO_2 radical (R19). The further decomposition of HO_2 leads to the formation of OH radical through R20 or the reaction sequence of $HO_2 \rightarrow H_2O_2 \rightarrow 2OH$. These pathways contribute 75% to the OH formation. Thus, the reaction between sC_4H_8OH-1 and O_2 is a key one in the sC_4H_9OH chemistry especially at low temperature. The decomposition temperature of 2-butanol is 100 K higher than the experimental results without considering R18; and the first peaks of 1-C_4H_8, H_2O, CO and C_2H_4 then disappear over the temperature range of 800–900 K.

$$sC_4H_9OH + OH = sC_4H_8OH\text{-}1 + H_2O \quad \text{(R19)}$$

$$CH_3 + O_2 = CH_2O + OH \quad \text{(R20)}$$

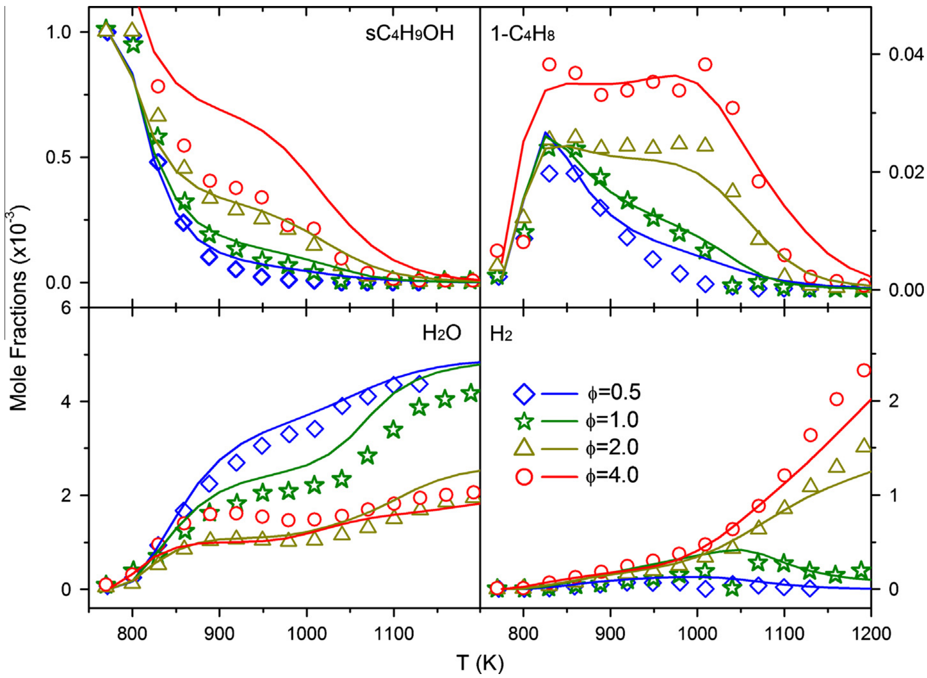

Fig. 20. Measured [11] (symbols) and simulated (solid lines) mole fraction profiles of sC_4H_9OH, 1-C_4H_8, H_2O and H_2 from 2-butanol oxidation in jet-stirred reactor at 10 atm with four equivalence ratios and $\tau = 0.7$ s. Carbon balance in these experiments is typically 100 ± 10%.

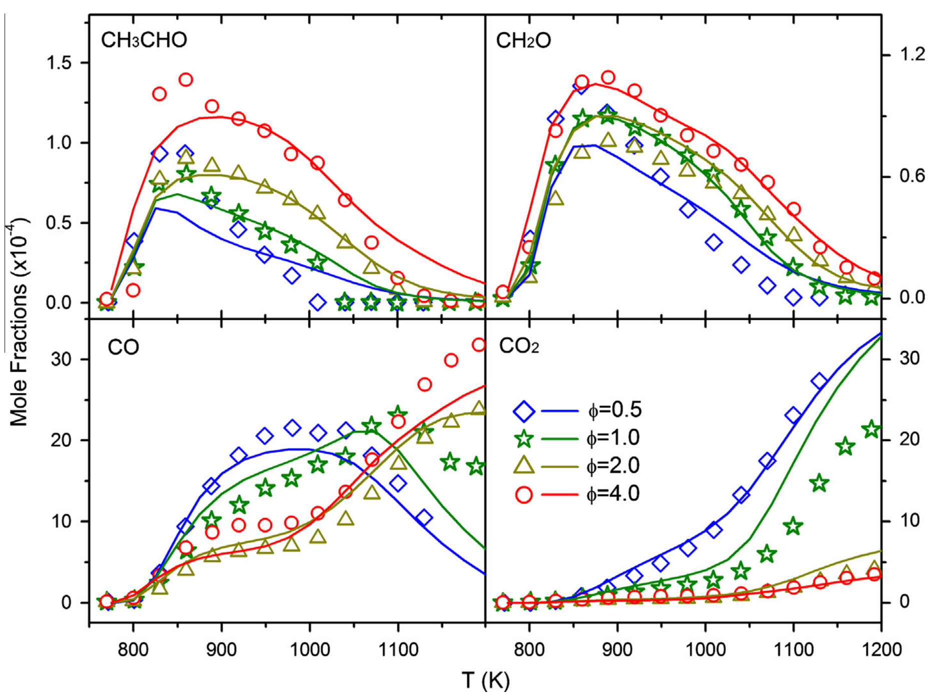

Fig. 21. Measured [11] (symbols) and simulated (solid lines) mole fraction profiles of CH_3CHO, CH_2O, CO and CO_2 from 2-butanol oxidation in jet-stirred reactor at 10 atm with four equivalence ratios and $\tau = 0.7$ s. Carbon balance in these experiments is typically 100 ± 10%.

The reaction pathways at 1025 K are quite different from those at 825 K. The ROP analysis at 1025 K shows that approximately 27% of 2-butanol decomposes through unimolecular reactions (R1)–(R6), while these reactions contribute less than 1% at 825 K. Approximately 40% of 1-C_4H_8 and 2-C_4H_8 are produced by the H_2O elimination reactions from 2-butanol (R1)–(R3). The ROP analysis also shows that at 1025 K, sC_4H_8OH-1 is dominated by β-scission reactions (R10, 87% and R11, 7%), which is the most distinguished feature between 825 K and 1025 K. HO_2 radical mainly originates from the reaction between O_2 and small radicals such as H, HCO, C_2H_5 and sC_2H_4OH.

The comparison of the simulated results by the present model and previous models at $\phi = 1.0$ and $\phi = 4.0$ can be seen in Figs. S5a and S6a in the Supplementary materials. In Sarathy

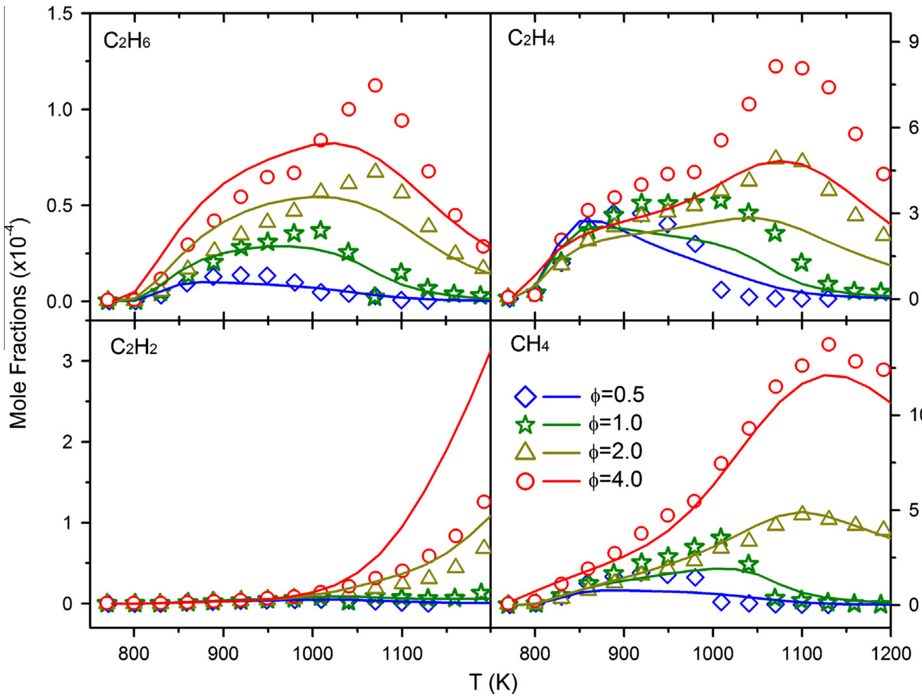

Fig. 22. Measured [11] (symbols) and simulated (solid lines) mole fraction profiles of C_2H_6, C_2H_4, C_2H_2 and CH_4 from 2-butanol oxidation in jet-stirred reactor at 10 atm with four equivalence ratios and τ = 0.7 s. Carbon balance in these experiments is typically 100 ± 10%.

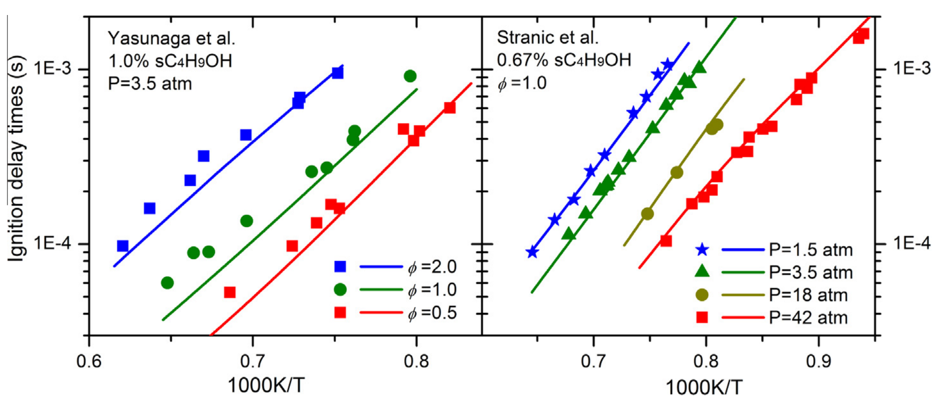

Fig. 23. Ignition delay times of 2-butanol measured by Yasunaga et al. [10] and Stranic et al. [9] (symbols) and simulated results (lines) by the present model.

et al. model [16], the predicted mole fraction of 1-C_4H_8 is under-predicted by a factor of 4 at ϕ = 1.0 and ϕ = 4.0. The most likely reason for this is that both reactions R1 and R16 were under-estimated in Sarathy et al. model [16]. The maximum mole fraction of 1-C_4H_8 predicted by Grana et al. model [17] was under-predicted by a factor of 3 at ϕ = 1.0 and 825 K, but over-predicted by a factor of 2 at ϕ = 4.0 and 1025 K, which indicates that in their model the rate constant of reaction R1 was over-estimated but the rate constant of R16 was far more under-predicted in their model.

The mole fractions of CH_2O and CH_3CHO reach its maximum at near 860 K, as shown in Fig. 21. The present model can predict the mole fraction profiles of key intermediates in JSR oxidation within the experimental uncertainties. The ROP analysis at 875 K shows that approximately 61% of CH_2O production is formed from the decomposition of CH_3O (R21), which is dominantly formed via the reaction between CH_3 and HO_2. The β-scission of C_2H_5O (11%) and the reaction between C_2H_3 and O_2 (9%) also contribute to the formation of CH_2O at 875 K.

$$CH_2O + H(+M) = CH_3O(+M) \quad \text{(R21)}$$

$$sC_2H_4OH + O_2 = CH_3CHO + HO_2 \quad \text{(R22)}$$

About 33% of CH_3CHO is produced from the reaction between sC_2H_4OH and O_2 (R22) at 875 K, and sC_2H_4OH is mainly produced from the β-scission of sC_4H_8OH-3. The tautomerization reaction between C_2H_3OH and CH_3CHO also contributes to the formation of CH_3CHO. A comparison of the simulation results of CH_2O and CH_3CHO between the present model and previous models at ϕ = 1.0 and ϕ = 4.0 is also included in the Supplementary materials (Figs. S5 and S6).

4.5. Ignition delay times

The ignition delay times of 2-butanol behind reflected shock tube waves have been measured by Moss et al. [8], Stranic et al. [9] and Yasunaga et al. [10]. The present model is validated against the ignition measurements by running the Constant Volume

Homogeneous Batch reactor in the Chemkin-Pro program. The ignition delay times were determined as the point of maximum temperature rising, when the OH concentration reaches half of its maximum.

Moss et al. [8] measured the ignition delay times of 2-butanol with the equivalence ratios varying from 0.25 to 1.0. They also changed the initial 2-butanol concentrations from 0.5% to 1.0% and pressures from 1.3 to 4.0 bar. Later, Stranic et al. [9] repeated the 2-butanol ignition-delay-time experiments under the same conditions. However, there is disagreement between the two sets of experiments. Stranic et al. [9] also measured the 2-butanol ignition delay times over the wide range of pressures (1.5–45 atm). Figure 23 shows that the data measured by Stranic et al. [9] is well predicted by the present model. Recently, Yasunaga et al. [10] measured the ignition delay times of 2-butanol with equivalence ratio varying from 0.5 to 2.0 at 3.5 atm. This data is also simulated by the present model and is illustrated in Fig. 23. It is concluded that the present model can accurately predict these sets of data with different pressures, equivalence ratios and initial 2-butanol concentrations. Furthermore, the present model can well predict the flame speeds measured by Veloo and Egolfopoulos [7], which is provide in Fig. S8 of the Supplementary materials.

5. Conclusion

The pyrolysis of 2-butanol in the flow reactor at pressures of 5, 30, 150 and 760 Torr were studied using SVUV/PIMS over the temperature range of 900–1550 K. About 25 pyrolysis species were identified in the experiment. Their mole fraction profiles versus the pyrolysis temperatures and pressures were determined. The rate constants of unimolecular decomposition reactions of 2-butanol at various pressures were calculated by using the RRKM/Master Equation method. With the help of theoretical calculations, a detailed kinetic model consisting of 160 species and 1038 reactions was developed to simulate the pyrolysis experiments. Generally, the present model can predict most of the pyrolysis species within the experimental uncertainty. The rate of production analysis was performed to analyze the reaction pathways of 2-butanol pyrolysis. It is concluded that unimolecular reactions play significant roles in the 2-butanol pyrolysis. Sensitivity analysis shows that the simulated mole fractions of primary products are very sensitive to the rate constants of those unimolecular reactions. The calculated rate constants of the unimolecular reactions are well validated by the pyrolysis experiments.

Furthermore, the present model was validated by the species profiles in shock tube pyrolysis among the pressure range of 1.3–11.9 atm, the rich laminar premixed 2-butanol flame at 30 Torr and oxidation experiments in jet-stirred reactor with equivalence ratios varying from 0.5 to 4.0 at 10 atm. The model was also validated by the ignition delay times with pressures ranging from 1 to 45 atm and equivalence ratios changing from 0.25 to 2.0. Good agreement was obtained between the simulated and the experimental results of pyrolysis, oxidation in JSR, ignition delay times and laminar flame speed. It is indicated that the present 2-butanol model is suitable for wide range conditions with the temperature ranging from of 800 to 2200 K, the pressure varying from 5 to 7600 Torr, and the equivalence ratio changing from of 0.25 to pyrolysis conditions.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2013.04.010.

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